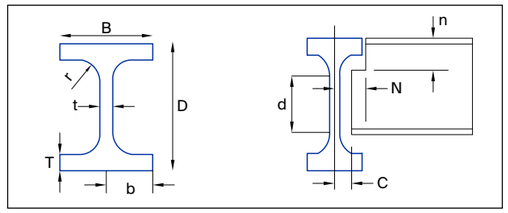
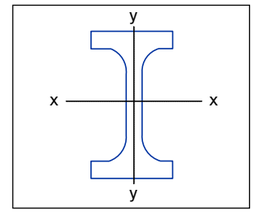
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W12-12x12 (305x305)	190	282.9	365.3	322.2	44.1	26.8	15.2	246.7	360	3.65	9.21	15	158	59
	210	312.5	373.6	324.9	48.3	30.0	15.2	246.7	399	3.37	8.23	17	157	63
	230	342	382.3	327.5	52.6	32.6	15.2	246.7	437	3.11	7.57	18	157	68
	252	375	391.4	330.3	57.2	35.4	15.2	246.6	478	2.89	6.97	20	157	72
	278	413.7	402.6	333.8	62.7	38.9	15.2	246.7	529	2.66	6.35	21	157	78
	279	415	402.6	333.8	62.7	38.9	15.2	246.8	528	2.66	6.34	21	157	78
	305	454	414.5	336.2	68.7	41.3	15.2	246.7	578	2.45	5.97	23	157	84
	336	500	427.2	340.0	75.1	45.1	15.2	246.6	638	2.26	5.47	25	157	90
W14-14x5 (356x127)	22	33.1	349.0	125.4	8.5	6.0	10.2	311.6	42.1	7.38	51.9	5	70	19
	26	39.1	353.4	126.0	10.7	6.6	10.2	311.6	49.8	5.89	47.2	5	70	21
W14-14x6 3/4 (356x171)	30	45.0	351.4	171.1	9.7	7.0	10.2	311.6	57.3	8.82	44.5	6	92	20
	34	51.0	355.0	171.5	11.5	7.4	10.2	311.6	64.9	7.46	42.1	6	92	22
	38	57.0	358.0	172.2	13.0	8.1	10.2	311.6	72.6	6.62	38.5	6	92	23
	45	67.1	363.4	173.2	15.7	9.1	10.2	311.6	85.5	5.52	34.2	7	92	26
W14-14x8 (356x203)	43	64.0	347.0	203.1	13.4	7.8	15.2	289.8	81.2	7.58	37.4	6	108	29
	48	71.4	350.3	204.0	15.1	8.6	15.2	289.7	91.2	6.75	33.5	6	108	30
	53	78.9	353.6	204.7	16.7	9.4	15.2	289.8	100	6.13	30.8	7	108	32
W14-14x10 (356x254)	61	90.8	352.8	253.9	16.3	9.5	15.2	289.8	115	7.79	30.4	7	132	32
	68	101.2	356.6	254.8	18.3	10.5	15.2	289.6	129	6.97	27.5	7	132	33
	74	110.1	359.9	255.8	19.9	11.4	15.2	289.6	141	6.41	25.3	8	132	35
	82	122	363.5	257.3	21.7	12.9	15.2	289.7	155	5.92	22.5	8	132	37
W14-14x14 1/2 (356x368)	87	129	355.6	368.6	17.5	10.4	15.2	290.2	164	10.5	27.9	7	189	33
	90	133.9	356.1	368.8	18.0	11.2	15.2	289.6	171	10.2	25.9	8	189	33
	99	147.3	359.7	370.0	19.8	12.3	15.2	289.7	188	9.34	23.5	8	189	35
	103	152.9	362.0	370.5	20.7	12.3	15.2	290.2	195	8.95	23.6	8	189	36
	109	162.2	363.7	371.0	21.8	13.3	15.2	289.6	207	8.49	21.7	9	189	37
	119	177	368.2	372.6	23.8	14.4	15.2	290.2	226	7.83	20.2	9	189	39
	120	178.6	367.8	372.6	23.9	15.0	15.2	289.6	228	7.80	19.3	9	189	39
	132	196.4	372.4	374.0	26.2	16.4	15.2	289.7	250	7.15	17.7	10	189	41
	136	201.9	374.6	374.7	27.0	16.5	15.2	290.2	257	6.94	17.6	10	189	42
W14-14x16 (356x406)	118	175.6	360.0	396.0	20.0	20.0	15.0	290.0	224	9.90	14.5	12	198	35
	145	215.8	375.4	393.7	27.7	17.3	15.2	289.6	275	7.11	16.8	11	198	43
	158	235.1	381.0	394.8	30.2	18.4	15.2	290.2	299	6.54	15.8	11	198	45
	159	236.6	380.5	395.4	30.2	18.9	15.2	289.6	302	6.54	15.3	11	198	45

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.94	78870	24640	14.8	8.27	4318	1529	5105	2342	0.855	7.7	6.35	2034
1.96	89270	27660	15.0	8.33	4779	1702	5703	2613	0.855	7.1	7.32	2693
1.98	100600	30880	15.2	8.41	5261	1886	6332	2898	0.856	6.6	8.39	3490
2.01	113200	34470	15.4	8.49	5783	2087	7020	3211	0.856	6.2	9.62	4495
2.04	129600	39040	15.7	8.59	6437	2339	7890	3604	0.857	5.7	11.3	5953
2.04	129500	39010	15.7	8.59	6435	2338	7888	3602	0.857	5.7	11.3	5946
2.07	147600	43690	16.0	8.69	7122	2599	8807	4006	0.860	5.4	13.1	7719
2.10	169000	49420	16.3	8.80	7913	2907	9882	4487	0.861	5.0	15.3	10110
1.17	8250	280	14.0	2.58	473	44.7	543	70.3	0.863	42.2	0.081	8.79
1.18	10170	358	14.3	2.68	576	56.8	659	89.1	0.871	35.2	0.105	15.1
1.36	12070	811	14.5	3.76	687	94.8	775	147	0.874	36.8	0.237	15.8
1.36	14140	968	14.8	3.86	796	113	896	174	0.881	32.1	0.286	23.8
1.37	16040	1108	14.9	3.91	896	129	1010	199	0.882	28.8	0.330	33.4
1.38	19460	1362	15.1	3.99	1071	157	1211	243	0.886	24.4	0.412	55.7
1.46	17760	1874	14.8	4.80	1024	185	1138	283	0.891	25.9	0.521	43.2
1.47	20170	2140	14.9	4.84	1151	210	1285	322	0.891	23.3	0.601	60.4
1.48	22480	2391	15.0	4.88	1271	234	1424	359	0.892	21.3	0.679	80.1
1.68	26540	4451	15.2	6.21	1505	351	1668	534	0.886	21.5	1.26	90.3
1.69	30060	5048	15.3	6.26	1686	396	1878	604	0.886	19.4	1.44	126
1.69	33120	5569	15.3	6.29	1840	435	2058	665	0.886	18.0	1.61	161
1.70	36680	6175	15.4	6.31	2018	480	2270	734	0.884	16.6	1.80	211
2.14	40250	14610	15.6	9.43	2264	793	2479	1199	0.844	19.9	4.18	153
2.14	41580	15080	15.6	9.40	2335	818	2565	1238	0.842	19.2	4.31	169
2.15	46240	16730	15.7	9.43	2571	904	2838	1370	0.842	17.7	4.83	223
2.16	48590	17550	15.8	9.49	2685	948	2965	1435	0.844	17.0	5.11	251
2.16	51540	18600	15.8	9.49	2834	1003	3142	1519	0.843	16.2	5.43	296
2.17	57120	20530	15.9	9.54	3103	1102	3455	1671	0.844	15.0	6.09	381
2.17	57290	20600	15.9	9.51	3115	1106	3475	1678	0.842	14.9	6.09	390
2.18	63720	22820	16.0	9.55	3422	1221	3838	1853	0.843	13.7	6.84	511
2.19	66260	23690	16.1	9.60	3538	1264	3972	1920	0.844	13.4	7.16	558
2.24	51770	20730	15.2	9.61	2876	1047	3235	1603	0.810	16.1	5.99	322
2.26	71240	28180	16.1	10.1	3796	1432	4264	2172	0.832	13.0	8.52	632
2.28	79090	30990	16.3	10.2	4151	1570	4687	2383	0.834	12.1	9.54	812
2.28	79160	31170	16.2	10.2	4161	1577	4702	2394	0.833	12.0	9.56	822